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Project Title:	Gravitational Influence on Cerebellar Control of Gaze Movement and Adaptation		
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Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 5/2009)		
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COI Name (Institution):	Fuchs, Albert (University of Washington) Robinson, Farrel (University of Washington)		
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Task Description:	Micro-gravity alters the normal relationships between coordinated multi-axial body movements and sensory feedback. As subjects undergo transitions between 1G to near 0G, they experience inappropriate eye and head movement and disorientation when shifting gaze. These changes pose a substantial challenge to mission objectives. One effort to address such challenges has been to develop context dependent preadaptation training protocols to train astronauts to quickly adapt based on inertial cues. Unfortunately, while this approach holds much promise, little is known about the reference frame for such adaptation or the underlying physiological mechanisms responsible for these adaptive changes. This black box approach limits progress toward the development of appropriate countermeasures. Our approach is to develop a laboratory analog of the altered gravitational/inertial environments, and then use identical psychophysical tests to evaluate behavioral performance of human and animal subjects in these environments to define both the neural mechanism and the crucial factors determining adaptation. Our research will consist of 3 components. First, we will describe the characteristics of context dependent human adaptation and its transfer to other forms of adaptation, seeking to localize behaviorally the site of adaptation in the CNS. Second, we will perform identical experiments on rhesus monkeys while recording the neural activity of cells within the caudal fastigial nucleus, which is thought to be important for saccadic adaptation. Third, we will simulate changes in the vestibular environment by changing the activity of the vestibular end organ in monkeys. We will study context dependent adaptation and the neural discharge associated with adaptation in this altered vestibular environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	

We have made continued progress on each aspect of our proposed research project in the second year of approval and we are still on target to complete the entire project on schedule. Our progress this year will result in 1 poster presentation and 4 submitted manuscripts. Each group working on this project has contributed significant findings. In addition, in related activity, we have advanced our development of a new technology for eye and head movement recording with the submission of an R-21 research application to the NIDCD-NEI. Furthermore, our collaboration with the group of Dr. Millard Reschke at Johnson Space Center, Houston, on the interpretation of human saccadic eye movement data collected before, during, and immediately following spaceflight, has resulted in a manuscript and a commitment for further collaboration on the interpretation of other existing data. Our progress for the second year is documented below.
Human Experiment - Year 2 update. (Fuchs, Ling, Hopp, and Phillips) – Our current results confirm that eye and head movements are adapted together during head unrestrained gaze adaptation in humans. This data was reported at the Society for Neuroscience 35th Annual Meeting, 2005, in Washington, DC. We have also shown that adaptation of human saccadic eye movements in the head restrained laboratory condition transfers to the head unrestrained condition, and adaptation of head unrestrained gaze movements transfers to the head restrained condition. Furthermore, the transfer shows the same "adaptation field" in both the head unrestrained and head restrained conditions. This result has important operational significance because it suggests that a human preflight saccade adaptation countermeasure will transfer to the context in which natural head unrestrained gaze movements are being performed. Such transfer is critical to any useful countermeasure, since astronauts will not be performing saccadic eye movements under laboratory conditions during 0G to 1 G transitions, but rather will be performing natural head unrestrained gaze shifts.

Human Experiment - New in Year 2. (Fuchs, Ling and Phillips) Our preliminary results suggest that changes in orientation including roll tilt and inversion of subjects have significant effects on the kinematics of saccadic and smooth eye movements in the horizontal or vertical plane. In particular, inversion of human subjects produces consistent decreases in saccade precision, and variable changes in accuracy. Smooth pursuit shows an increase in saccadic intrusions, and decreases in gain in many subjects during inverted viewing of sinusoidally moving targets. No consistent changes were observed in horizontal optokinetic nystagmus, although vertical OKN was affected. These results are interesting in that they imply that human eye movements in both the horizontal and vertical plane are affected by changes in whole body orientation.

Human Experiment - Year 2 Update. (Fuchs and Hopp) – This work has resulted in a manuscript. Our results confirm that cognitive volitional saccade adaptation transfers across conditions. For example, transfer occurs between memory guided and delayed saccades. The previous work suggested that human saccadic eye movement adaptation transfers between reflexive saccades, such as targeting saccades and express saccades. There is poor transfer between reflexive saccade adaptation and cognitive saccades. In addition, in year 2, we have confirmed that oblique saccades transfer along the vector of the adaptation and not along the individual components of the adaptation. These provide both a benefit and a challenge for operational implementation of a preflight adaptation countermeasure. We have evidence that there is transfer between adaptation of specific saccade types and other saccade types that were not specifically adapted in the laboratory but which would presumably be utilized during spaceflight. We also now know, however, that human saccade adaptation does not transfer as widely across saccade types as does non-human primate saccade adaptation. Also, adaptation of human saccades with a given horizontal or vertical component will not transfer to all saccades with components of comparable amplitude, but rather only to saccades with a comparable vector. Therefore, multiple preflight adaptations are likely to be required to serve as an effective countermeasure for saccades of different types, amplitudes and directions. While the precise specificity remains to be determined, we now know that component adaptation will not be an effective shortcut to the development of generalized adaptation states.

Animal Experiment - Year 2 update. (Phillips, Fuchs, Ling). This work has resulted in a manuscript. Elimination of canal afferent input modifies the reflex behavior and discharge of brainstem neurons during head unrestrained gaze shifts in rhesus monkeys. We trained 2 rhesus monkeys to perform head unrestrained gaze shifts and then lesioned the vestibular end organ with the use of bilateral canal plugging. We are currently recording from a 3rd monkey. Elimination of modulated vestibular input via this procedure results in a change in the coordination of eye and head movements during head unrestrained gaze shifts. We observed that animals made accurate gaze shifts following canal plugging, but the relationship of saccadic eye movement end and gaze movement end, and the amplitude and duration of the eye and head movement components was dramatically altered. This data suggests that vestibular signals are important for the normal control of gaze movement, but that new strategies emerge, involving adaptation of saccade amplitude, to compensate for the loss of vestibular input and maintain accurate gaze movements after canal plugging. Furthermore, we have recorded cells in the brainstem burst generator and vestibular nuclei before and following bilateral canal plugging.

The results of burst generator recording suggest that vestibular signals converge at both the premotor neuron and motoneuron level to control gaze movements in normal monkeys, and that the burst generator loses the capacity to drive

an accurate gaze movement to completion following loss of vestibular input. Other mechanisms, downstream of the pontine burst generator, a putative target of caudal fastigial nucleus inputs, are required to restore accurate gaze movement following vestibular loss. The preliminary results of vestibular nucleus recording suggest that there is not an emergence of an efference copy signal at the level of the vestibular nuclei following vestibular loss, eliminating one potential site for this downstream mechanism. These data are critical for assessing the importance of vestibular signals during gaze shifts with normal and abnormal vestibular contributions. Our data suggests that disruption of vestibular inputs does not prevent the successful adaptation of saccades and the emergence of new control strategies for accurate gaze. The site of these adaptive changes remains elusive.

Animal Experiment. Year 2 update (Fuchs, Ling, and Phillips) We have continued to analyze and model data recorded from a subset of neurons in the caudal fastigial nucleus (CFN) of 2 rhesus monkeys that display discharge characteristics consistent with a head movement signal during active head unrestrained gaze shifts. This discharge is similar to the head related signal of prolonged burst neurons recorded in the pontine brainstem during head unrestrained gaze shifts, in that it dramatically outlasts the saccade related discharge of saccade neurons in the CFN. Such a neuron could carry the vestibular signals requisite for the control of coordinated gaze movement during head unrestrained gaze shifts. Likewise, these cells are possible targets for the vestibular inputs that may be required to establish the context for context dependent adaptation. We have just begun to record from this population during both head unrestrained gaze shifts and tilt context dependent adaptation (22 cells in 1 monkey in year 2).

Task Progress:

Animal Experiment. Year 2 Update (Noto, Robinson and Phillips) This work has resulted in a manuscript. In rhesus monkeys, saccades made during tilt orientations in pitch and roll are dissymmetric, producing dissociated vertical deviation of the eye and, in some monkeys, changes in saccade velocity and amplitude. This data suggests that vertical skew deviation does accompany roll tilt of approximately 45 degrees in monkeys during viewing of near point visual stimuli. This result is similar to the results obtained by Betts et al, 1995 in humans, and contrasts with the absence of skew deviation seen in earlier studies of non-human primates. We recorded the saccades of 3 rhesus monkeys during 45 deg static tilt in pitch and roll and observed the resulting targeting saccades. The saccades were observed to be dissymmetric in all animals, with a vertical deviation of the ipsilateral eye upward and the contralateral eye downward during static roll tilt downward. This result suggests that a vertical diplopia exists during roll tilt in rhesus monkeys which is uncompensated by normal short-term vestibulo-ocular mechanisms. This could pose a significant challenge to mission objectives if similar dissociations are present in humans under similar conditions, and if the diplopia compromises vision under some visual stimulus conditions, i.e., reduced lighting or required fixation of weakly illuminated avionics during disorienting 1G to 0G transitions.

Animal experiment. Year 2 update. (Phillips, Noto and Robinson) We determined, in rhesus monkeys, that the adaptation field during tilt is in craniotopic coordinates and is largely unaffected by tilt orientation. Adaptation of saccade gain is restricted to a limited range of vectors relative to the head, which are directionally maintained during saccade testing in other orientations. We tested the saccades of rhesus monkeys to 10-degree target steps in 8 cardinal directions, and then adapted rightward saccades so that they had reduced gain. We then retested the saccades in 8 directions during roll tilt to the right or left, showing a constant adaptation field relative to the head. This data was reported at the Society for Neuroscience 34th Annual Meeting, 2004, in San Diego, CA. In addition, we have found a similar phenomenon in human subjects subjected to the same testing paradigm. This confirms an important limitation to preflight adaptation paradigms as a countermeasure. Since the adaptation is restricted to specific vectors with respect to the head, the range of vector directions must be extended during preflight training to allow for adequate representation of visual space during natural movements.

Animal experiment - Year 2 update. (Phillips, Noto and Robinson). Rhesus monkeys display transfer of adaptation across tilt adaptations, and initial tilt orientation does affect to degree of transfer between orientations. In these experiments, we produced a gain decrease adaptation in one direction along the horizontal meridian with respect to the head at a specific tilt orientation (pitched forward, pitched back, roll right, roll left or upright) and then changed the tilt orientation to another orientation to observe transfer of adaptation. Adaptation transferred to each tilt orientation, with the best transfer occurring when the initial adaptation was in the normal, head upright orientation. This data was reported at the Society for Neuroscience 35th Annual Meeting, 2005, in Washington, DC. Also, the rate constants for adaptation were faster in the head upright condition. We have also begun studies on the transfer of adaptation in human subjects in year two. These studies are not complete, but should be completed by the end of year two. If this adaptation transfers in humans, adaptation-training countermeasures will transfer to a range of tilt orientations or inertial contexts.

Animal Experiment - New in year 2. (Phillips, Noto, Ibiretta and Robinson). Context dependent adaptation has been performed in rhesus monkeys during pitch and roll tilt. In these experiments, animals are forward adapted in one tilt adaptation and back adapted in a second tilt adaptation. Our results indicate that there is generalized adaptation independent of context. Since back adaptation is stronger than forward adaptation, animals show a reduction of gain in the combined context adaptation experiments independent of tilt context. This is true of pitch or roll contexts. However, context cues do produce differences in adaptation in roll tilt, but not in pitch tilt. This suggests that gravitational cues are not equivalent for tilt context adaptation. Rather, roll cues are more salient than pitch cues. The precise mechanism for this remains to be elucidated. However, since roll tilt produces differential ocular torsion in the two eyes and pitch cues do not, it is possible that efference copy or reafference from the oculomotor system is the critical factor for context dependent saccade adaptation; i.e., the oculomotor torsion provides the context and not the vestibular input per se. This result may have profound implications for the application of context dependent saccade adaptation paradigms as a countermeasure, since the implication is that the context is provided only secondarily by vestibular input - the primary cue is oculomotor. This data was reported at the Society for Neuroscience 35th Annual Meeting, 2005, in Washington, DC.

Animal Experiment - New in year 2. (Phillips, Noto, Andrews, and Robinson). We have begun recording vestibular and saccadic responses in the CFN of monkeys that are subjected to roll tilt. Our results are largely consistent with the results of Shaikh et al, 2005 in that cells that burst for saccades, saccade cells, do not appear to carry passive head rotation signals. However, we have also recorded from saccade cells during saccades in different tilt orientations. Such cells do show small but significant changes in their discharge for saccades during roll tilt, despite the absence of a response to passive rotation. Further, in two cells, we have recorded changes in saccade related discharge for saccades made during passive head rotation; i.e., ipsi-versive saccades are associated with a different discharge than contraversive saccades. This very preliminary result is intriguing, and suggests that the CFN may be responsible for the coordination of vestibular and saccadic eye movement commands even in response to passive vestibular stimuli. These results are important because they indicate that we may be on the right track in our search for the neural mechanism underlying changes in eye movements associated with transition from 1.0G to near 0.0 G.

	Related activity. (Kaneko, Phillips) As an extension of our year one activity in which we began development with a local firm of a printed circuit device that could serve as a cheap alternative to the existing eye movement transducers, in year 2 we have applied for an R21 technology development grant to continue the development of this exciting technology. Our proposal to the NIH, which is pending, is to test this technology in animals with an objective of developing a viable tool both for human and animal use. The continued development of this technology is a direct outgrowth of our experimental research funded through this project. Related activity. (Reschke, Phillips) This work has resulted in a manuscript. Our ground based research compliments flight research begun many years ago at Johnson Space Center by Reschke, Harm, Paloski, Krnavek, Somers, and Ford. Dr. Reschke recorded pre-flight, flight and post-flight saccadic eye movement data in astronauts during several space missions. We have analyzed this very important flight data to better understand the nature of eye movement changes related to adaptation to novel inertial environments. Specifically, we have analyzed the metrics of saccades and the stability of post saccadic gaze. It is critical that we understand nature of the changes in gaze movement control during the transition to and from 1 G to near 0 G to optimize strategies for applying proposed countermeasures.
Bibliography Type:	Description: (Last Updated: 05/13/2009)
Abstracts for Journals and Proceedings	Phillips JO, Noto CT, Iberreta M, Robinson FR. "Context Dependence of Human and Nonhuman Primate Saccade Adaptation." Society for Neuroscience, 35th Annual Meeting, Washington, DC, November 2005. Society for Neuroscience, 35th Annual Meeting, abstract 859.7, 2005. , Nov-2005